

Overview

In this mini project you will dive into the world of SQL through creating and populating a database of your own and then extracting meaningful insights through SQL using queries and Python visualization techniques. You have the freedom to choose your own data, the sources it originates from, and the business case you wish to cater to.

Setup

In this Mini Project you will work in pairs, and are required to organize how to divide tasks throughout the week.

Brief

In this project, you'll start by exploring the [Willy Wonka Chocolate Factory dataset](#). You will then formulate a business case or decision that needs to be made inside the factory related to the data available. The goal is solve a problem that the company is or may be facing.

You'll sketch out an Entity-Relationship Diagram (ERD) highlighting significant data fields and relationships, then proceed to create and populate a functional database. During the process, you will carry out data wrangling in Python and use SQL to run queries, derive valuable insights, and summarize your findings. SQL queries are not very good for visualizations so they should ideally be used to extract overall KPIs and metrics across a few regions of interest.

You will also use Python to complement your SQL findings through some data visualizations, encapsulating your data story from exploration to insights.

Note: You don't need to connect Python to SQL for the visualizations but instead can use the intermediate `.csv` files that you will create for the database schema.

Daily Tasks

Day 1: Project Initiation & Data Selection

- **Data Selection:** For this project, you will be using the [Willy Wonka Chocolate Factory dataset](#).**
- **Business Framing:** Clearly articulate the business challenge you're addressing. Craft hypotheses to guide your exploration.
- **Project Planning:** Map out your project's trajectory using tools like Jira or Trello.
- **Extract:** Examine the data and try to understand what the fields mean before building the database.

Day 2: Data Examination & Schema Design

- **Database Design & Creation:**
 - Engage with your data to discern the significance of each field.
 - Sketch an Entity-Relational-Model, highlighting primary keys, foreign keys, and table relationships.
 - Translate the ERD into a functional database.
 - Make sure to also define the proper data types for each column.
- **Transform:**
 - Do data wrangling to transform your dataset according to your analysis goals.
- **Load:**
 - Safely import the sanitized data into your database, upholding its structural and data integrity.

Day 3: SQL Queries & Analysis

- **SQL Queries and Insights:** use SQL commands to derive insights.
 - Leverage functionalities like **JOIN**, **GROUP BY**, **ORDER BY**, **CASE**, and subqueries.
 - Summarize the data using **mean**, **max**, **min**, **std** and more.

Day 4: Analysis and Visualization

- **Data Visualizations:** use Python libraries to create compelling visual representations of your findings.
- **Report Compilation and finalization:** Synthesize all the analysis and insights into a comprehensive report that narrates your data story.
- **Presentation:** Design a compelling presentation that encapsulates the essence of your project, challenges, insights, and outcomes (see more details below).

Deliverables

The main deliverable is a GitHub repository which should contain the following:

- A [README.md](#) file in the root of the repository to document the project containing useful information about the project. Anyone that reads the README should be able to understand the project without having to look through all of the files.
- Code and files that demonstrate the data pipeline coverage of acquisition, transformation, loading, analysis, and reporting. This includes:
 - Database: The exported **.sql** file should be included with the final schema.
 - Entity relationship diagram (ERD).
 - SQL Queries: A compilation of all SQL queries used during the project.

- Python files. Remember, when using Python the code must be put in wrapped functions.
- At least 1 Jupyter notebook containing the report in full with visualizations.
 - The report should be comprehensive and aesthetically pleasing, that narrates the story of your data in full. The report should combine text, clean code, meaningful outputs, and engaging visualizations to convey the derived insights and conclusions effectively.
 - Make sure to modularize your work. The Jupyter notebook report **must** be distinct from the code responsible for cleaning, acquiring, processing data, etc. Feel free to use multiple Jupyter notebooks, **.py** or **.sql** modules for this.
- URL of the slides for the presentation:
 - The presentation (see guidelines below) should summarize your insights and conclusions.
 - Remember to **include your ERD diagram**. You should explain it during the presentation, and defend the decisions made.

Minimum Requirements

- **Research Objectives:**
 - Define at least two clear research questions or problem statements.
 - Analyze data effectively to address these questions, ensuring the results are presented in an understandable and coherent manner.
- **Data Acquisition:**
 - Utilize a minimum of two distinct sources for your data, such as CSV files, web scraping, or APIs.
- **Database Structure:**
 - Design a database incorporating a minimum of three tables.
 - Make sure to apply suitable primary and foreign keys and delineate the relationships between tables.
- **Data Management:**
 - Efficiently import data into your database using the right tools and methods.
 - Clean, format, restructure, and categorize the data to maintain consistency and accuracy.
- **Data Analysis:**
 - Craft at least five insightful SQL queries using fundamental clauses like GROUP BY, JOIN, and Subqueries.
- **Visualization:**
 - Produce a minimum of two compelling visualizations using tools such as Matplotlib or Seaborn.

Advanced Features (optional)

- **Scalable Data Pipeline:**
 - Develop functions to form a complete pipeline, allowing the data model to be updated with fresh data. This should cover extraction, transformation, and loading. Ensure that your analysis can adapt to new datasets.
- **Data Quality Assurance:**
 - Implement auxiliary functions that scrutinize the database for quality issues during updates.
- **Data Integration:** Incorporate multiple data sources (at least 2). To ensure a cohesive analysis, keep the following in mind:
 - **Relevance and Complementarity:** Ensure that your chosen datasets are related and complement each other to build a rich narrative.
 - **Tip:** Initiate your project with a dataset or by scraping a website, and then find an API or another CSV file that can enhance the information from your primary source.
 - **Interconnectivity:** Your datasets should share a common ground that allows them to be interconnected, facilitating a unified analysis in your report.
 - **Tip:** Merging datasets can be a complex process. Ideally, the datasets should have at least one column with matching data to successfully create a comprehensive dataset with enriched information.
 - **Tip:** If merging proves to be too challenging, consider demonstrating the relationship between the datasets through visualizations.
- **Advanced SQL Techniques:**
 - Harness advanced SQL methodologies, such as views and window functions, to gain deeper insights from the data.
- **Interactive Visualizations:**
 - Craft a minimum of five engaging visualizations using tools like Plotly. Embed these visualizations into a dynamic dashboard utilizing platforms like Dash or Streamlit.
- **Performance Optimization:**
 - Recognize the significance of indexes in boosting database efficiency and formulate indexes to accelerate query executions.
 - Amplify database and query performance by examining query execution plans, making necessary alterations to either the database schema or the queries themselves.
- **Documentation:**
 - High level documentation explaining tables designed and focusing on update methods.

Coding Best practices

- **Modularize Code:**
 - For Python: Create reusable functions and classes, and place them in separate `.py` files. Import them into your notebooks or main scripts to maintain a clean and organized codebase.
 - For SQL: Use a local organization for `.sql` files.
- **Naming Conventions:** Use clear and descriptive names for variables, functions, and classes. The same applies for the database, tables and columns naming. Follow language-specific naming conventions to enhance readability.
- **Remove Unused Code:** Eliminate any unused imports, variables, or lines of code/queries. Keep the codebase clean and focused on what is necessary.
- **Comments and Documentation:** Add meaningful comments and documentation to explain complex or crucial parts of the code. It aids understanding and future maintenance.
- **Logical Organization:** Group related functionalities together and maintain a coherent flow within your scripts or notebooks.
- **Regular Refactoring:** Review and refactor the code periodically to remove redundancies, enhance efficiency, and ensure alignment with current best practices and project needs.

Rubric

In order to assess your project and ensure all requirements are met, a **rubric** will be used. This rubric is used to **evaluate your project** by your teaching staff but also to **communicate** what constitutes incomplete, acceptable and excellent performance across each of the learning outcomes for the project. Take some time to review the rubric that is in Student Portal and ask your teacher any questions about it if necessary.

Tips & Tricks for Success

- Break the project down into small steps. Many smaller, simple tasks are better than a single complicated one.
- Work through and review the lessons in class to consolidate knowledge.
- Add relevant code to your project each day to maintain progress throughout the week.
- Commit early, commit often and don't be afraid of doing something incorrectly because you can always roll back to a previous version.
- Name your commits well so your teammates know what everyone is working on.

The Presentation

You will present your mini project on Saturday morning to the class showcasing your project as well as a short presentation in the form of some slides. You should present your findings to the class in a visually appealing, easily understandable manner. The format of the presentation will be as follows:

- Talking with Slides: 7 minutes
- Q&A: 5 minutes

Presentation Structure

Feel free to present the project in as you feel best represents your work, but below you will find a suggested format for the presentation as a guide.

1. **Title Slide** (1 slide):
 - Project title.
 - Team members' names.
2. **Project Overview** (1 slides):
 - Briefly describe the chosen dataset and its origin.
 - Highlight the business problem and the hypotheses that guided your approach.
3. **Data Acquisition, Enrichment, and Examination** (1 slide):
 - Highlight the primary sources of data and any complementary datasets.
4. **Database Design & Data Transformation** (1 slide):
 - Provide a concise visualization or description of the Entity-Relational-Model.
5. **SQL Insights & Advanced Analysis** (1 slide):
 - Showcase one or two standout insights derived from advanced SQL queries.
 - Highlight any specific analyzes that were particularly challenging or revealing.
6. **Visualization & Key Insights** (2 slides):
 - Show some of the primary visualizations developed.
 - Discuss the major insights derived from the visualizations.
7. **Conclusions & Business Implications** (1-2 slides):
 - Reflect on the original business problem and whether your findings support or refute your initial hypotheses.
 - Discuss the potential business implications of your findings.
8. **Closing Slide** (1 slide):
 - Reiterate the project title.
 - Team members' names.
 - A "Thank You" note or any final thoughts.

Total: Approximately 10 slides.

Famous Datasets

Below is a list of 48 famous datasets suitable for your project.

Dataset Name	Source URL	Description
Boston Housing Dataset	Kaggle Boston Housing	Information about the housing values in suburbs of Boston.
FIFA World Cup Players	Kaggle FIFA 2018	Player statistics from the FIFA World Cup 2018.
Supermarket Sales	Kaggle Supermarket Sales	Sales data for a supermarket chain.
Google Play Store Apps	Kaggle Google Play Store	Data about apps on Google Play Store.
US Elections Data	Harvard Dataverse US Elections	Comprehensive data on the US elections.
Spotify Music Data	Kaggle Spotify	Dataset of top tracks on Spotify.
World Happiness Report	Kaggle World Happiness	Survey data about the state of global happiness.
Credit Card Fraud Detection	Kaggle Credit Card Fraud	Dataset for credit card fraud detection problems.
Wine Quality	UCI ML Repository	Physicochemical data and quality ratings of wines.
Zomato Bangalore Restaurants	Kaggle Zomato	Details about restaurants in Bangalore listed on Zomato.
New York City Airbnb Open Data	Kaggle Airbnb NYC	Information about Airbnb listings in New York City.
Adult Income	UCI ML Repository	Predict whether income exceeds \$50K/yr based on census data.

Dataset Name	Source URL	Description
Bank Marketing	UCI ML Repository	Data related to marketing campaigns of a Portuguese bank.
Car Evaluation	UCI ML Repository	Evaluate the acceptability of a car based on various factors.
Human Resources Analytics	Kaggle HR Analytics	Data to analyze employee turnover.
Online Retail II	UCI ML Repository	A dataset containing all transactions occurring for an online retail company.
Bike Sharing Dataset	UCI ML Repository	Data about bike sharing systems from Washington DC.
Chess (King-Rook vs. King)	UCI ML Repository	Chess endgame database for predicting outcomes.
Forest Fires	UCI ML Repository	Predict the burned area of forest fires in Portugal.
Student Performance	UCI ML Repository	Factors affecting student achievements in secondary education.
Wholesale Customers	UCI ML Repository	Data on clients of a wholesale distributor.
Yeast	UCI ML Repository	Predicting protein localization sites in yeast.
Abalone	UCI ML Repository	Predict the age of abalone from physical measurements.
HTRU2	UCI ML Repository	Predicting a pulsar candidate using HTRU2 dataset.

Dataset Name	Source URL	Description
Ionosphere	UCI ML Repository	Classify radar returns from the ionosphere as good or bad.
Lenses	UCI ML Repository	Predictive data about the type of lenses that should be used.
Nursery	UCI ML Repository	Nursery dataset for decision making.
Page Blocks Classification	UCI ML Repository	Classify page blocks of documents as one of five classes.
Poker Hand	UCI ML Repository	Predicting poker hands from a dataset of drawn cards.
Seeds	UCI ML Repository	Measurements of geometrical properties of kernels belonging to different varieties of wheat.
Skin Segmentation	UCI ML Repository	Skin segmentation based on RGB values.
Wine	UCI ML Repository	Chemical analysis of wines grown in the same region in Italy.
Breast Cancer Wisconsin	UCI ML Repository	Diagnostic data for breast cancer cases.
Energy Efficiency	UCI ML Repository	Assessing energy efficiency of buildings.
Handwritten Digits (Optdigits)	UCI ML Repository	Image dataset of handwritten digits for recognition.
Heart Disease	UCI ML Repository	Dataset for predicting heart disease presence.

Dataset Name	Source URL	Description
Image Segmentation	UCI ML Repository	Image data for segmentation tasks.
Letter Recognition	UCI ML Repository	Identify letter characters from various fonts and styles.
Magic Gamma Telescope	UCI ML Repository	Data to simulate registration of high-energy gamma particles.
Mushroom	UCI ML Repository	Classify mushrooms as edible or poisonous.
Internet Advertisements	UCI ML Repository	Predict whether an image is an advertisement or not.
Libras Movement	UCI ML Repository	Classify movements in Libras (Brazilian sign language).
Lymphography	UCI ML Repository	Diagnostic data for lymphography patients.
EEG Eye State	UCI ML Repository	EEG data to determine the eye state.